

Metro Ticket Booking Digitization

Requirement Analysis Phase

Date	
Team ID	
Project Name	Smart Metro Ticket Booking System
Maximum Marks	4 Marks

2. Data Flow Diagram & User Stories

This document lays out the system's data flow at Level 0 (context) and Level 1 (process breakdown), followed by user stories that capture what each persona — Passengers, Station Managers, the Metro Operations Team, and IT Admins — needs from the system.

2.1 Data Flow Diagram — Level 0 (Context)

Data Flow Diagram — Level 0 (Context Diagram)

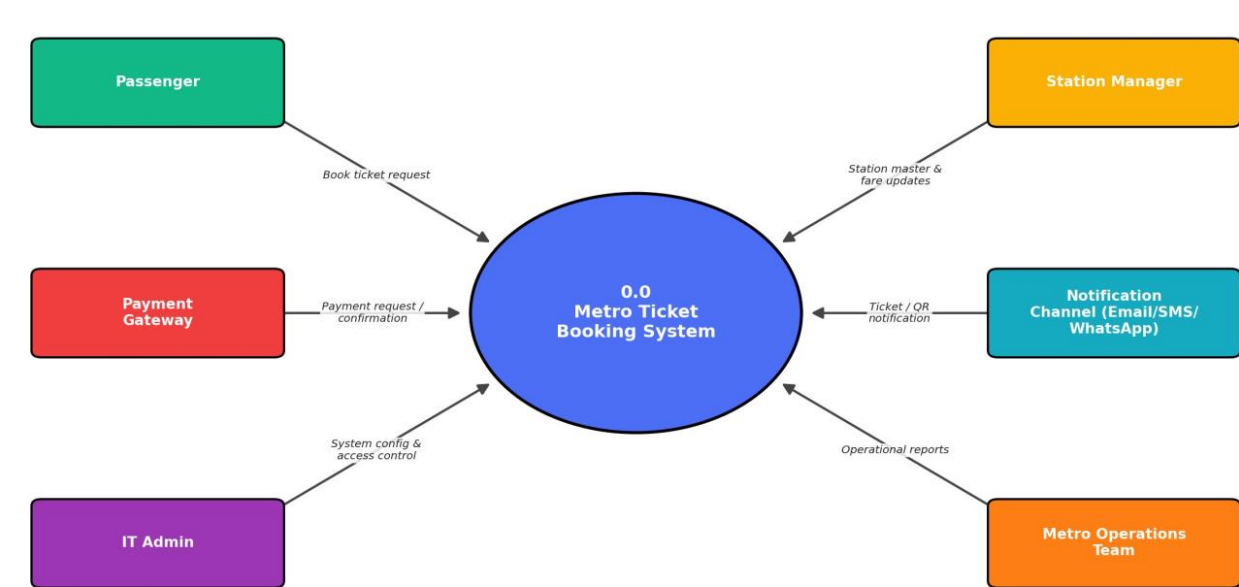


Figure — External entities interacting with the Metro Ticket Booking System

The context diagram shows four human actors and two external systems exchanging data with the core booking platform: Passengers start booking requests; Station Managers keep station and fare master data up to date; the Metro Operations Team consumes operational reports; IT Admins manage configuration and access; the Payment Gateway handles transactions; and the Notification Channel delivers tickets and alerts.

2.2 Data Flow Diagram — Level 1 (Process Decomposition)

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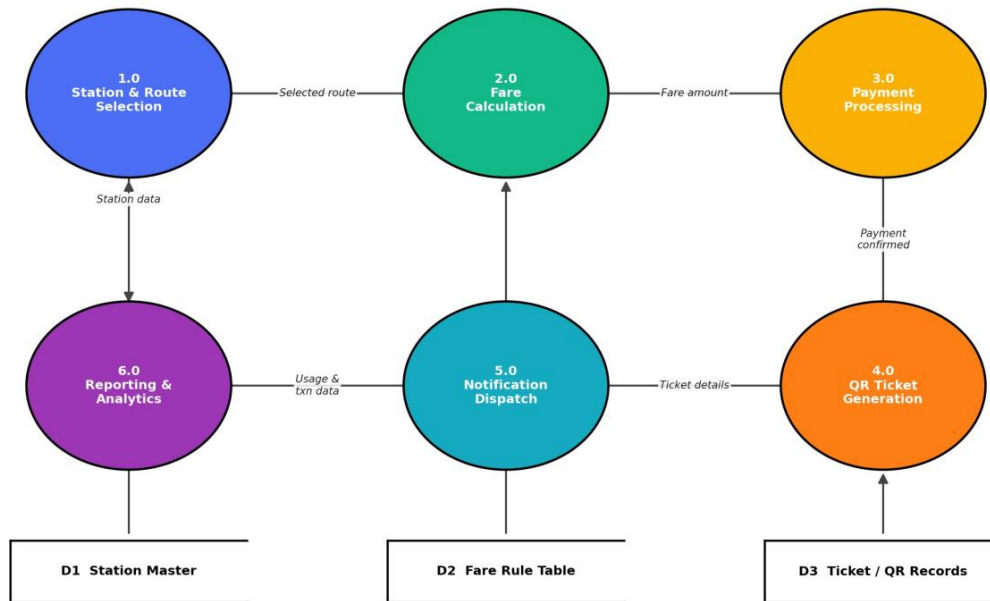


Figure — Decomposition of the booking system into six core processes

The system breaks down into six processes — Station & Route Selection, Fare Calculation, Payment Processing, QR Ticket Generation, Notification Dispatch, and Reporting & Analytics. These map directly onto Flow Designer sub-flows and Script Includes, which read from and write to three core data stores: the Station Master, the Fare Rule Table, and Ticket/QR Records.

2.3 User Stories

The user stories below are organized by persona, written as "As a <role>, I want <capability>, so that <benefit>", each paired with acceptance criteria.

Passenger (End User)

ID	User Story	Acceptance Criteria
US-01	As a passenger, I want to pick my source and destination stations from a searchable catalog form, so that I can book without visiting a counter.	The catalog form loads the station list from the Station Master; invalid source/destination pairs are blocked; search-as-you-type is supported.
US-02	As a passenger, I want to see the calculated fare right after entering my travel details, so that I know the exact amount before paying.	The fare updates live as station, passenger type, or ticket count changes; Student/Senior Citizen discounts apply automatically.
US-03	As a passenger, I want to pay digitally via UPI, card, or wallet, so that I don't need exact change.	The payment sub-flow supports at least three digital modes; failed payments show a clear retry option.
US-04	As a passenger, I want my ticket delivered as a QR code right after payment, so that I can travel without a printed ticket.	The QR is generated within seconds of payment confirmation and delivered via app notification and email/WhatsApp.
US-05	As a passenger, I want to view past bookings and re-download tickets, so that I have a record for reimbursement or reuse.	Booking history is available in the Service Portal, filterable by date and route.

Station Manager

ID	User Story	Acceptance Criteria
US-06	As a station manager, I want to maintain and update station details and fare zones, so that booking data stays accurate.	Station Master records are editable through a controlled catalog/table with an approval workflow for changes.
US-07	As a station manager, I want to validate QR tickets at the gate in real time, so that only paying passengers get through.	A QR scan returns a valid/invalid/used status within 2 seconds; duplicate scans are flagged.
US-08	As a station manager, I want a dashboard of today's ticket volume at my station, so that I can plan staffing and manage crowding.	The dashboard refreshes in near real time and can be filtered by hour.

Metro Operations Team

ID	User Story	Acceptance Criteria
US-09	As a metro operations manager, I want consolidated analytics on peak travel times and popular routes, so that I can plan schedules and resources.	Reports/Performance Analytics dashboards aggregate booking data by route, time band, and passenger type.
US-10	As a metro operations manager, I want visibility into revenue from digital payments versus cash, so that I can track digital adoption.	A revenue split report is available monthly and can be exported.
US-11	As a metro operations manager, I want automatic notifications for failed or disputed transactions, so that I can resolve issues proactively.	Flow Designer raises an incident/case record for any payment exception.

IT Admin

ID	User Story	Acceptance Criteria
US-12	As an IT admin, I want to configure catalog variables and fare rules without code changes, so that business changes can be made quickly.	The Fare Rule Table and catalog variables are admin-configurable through the ServiceNow UI.
US-13	As an IT admin, I want role-based access control across passenger, station manager, and operations views, so that data stays visible only to authorized users.	ACLs restrict table and UI Page access strictly by assigned role.
US-14	As an IT admin, I want monitoring and logs for the QR generation and payment integration, so that I can troubleshoot failures quickly.	Integration Hub spokes log request/response pairs with correlation IDs for traceability.